INCH-POUND

MIL-S-8805/20J
19 January 1993
SUPERSEDING
MIL-S-8805/20H
22 December 1989

MILITARY SPECIFICATION SHEET

SWITCHES, PUSH, MOMENTARY, 2 CIRCUIT, UNSEALED AND SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-S-8805

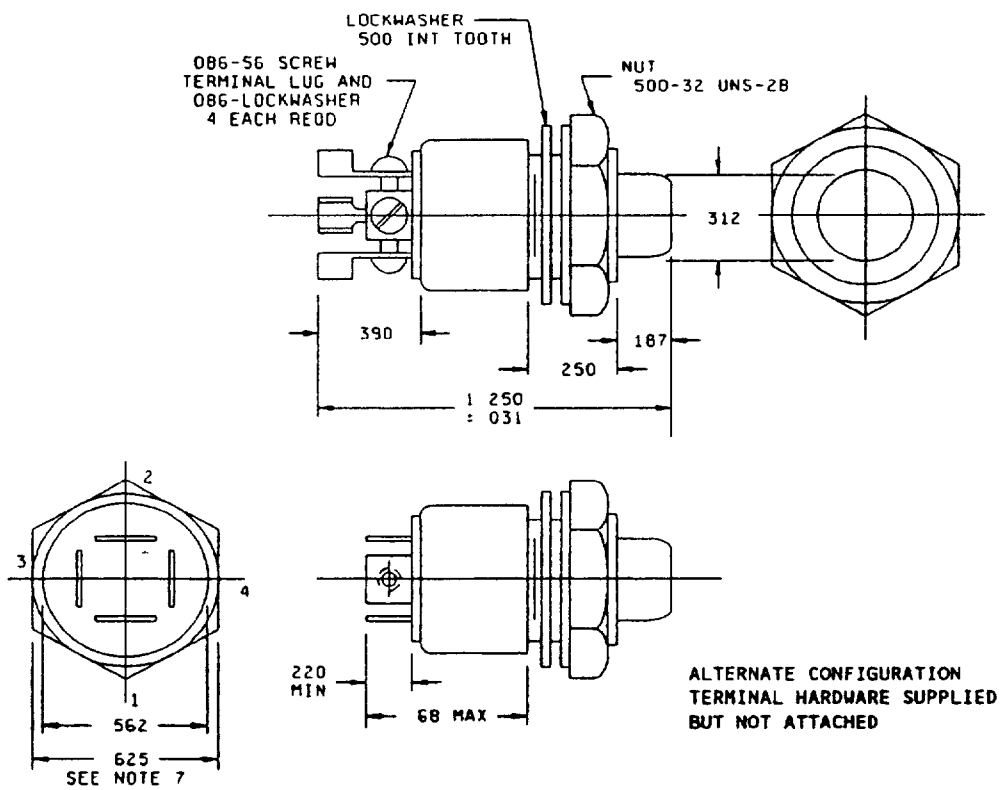
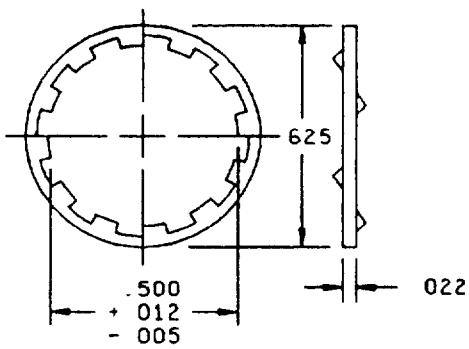


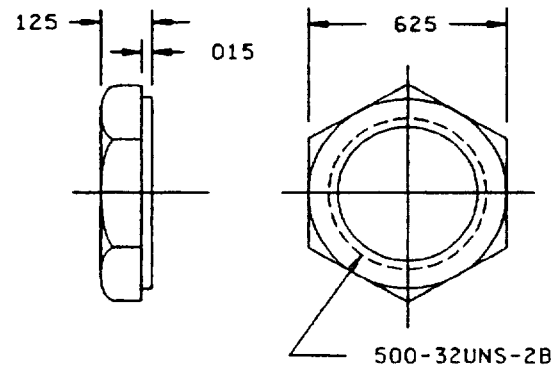
FIGURE 1 Configuration and dimensions

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

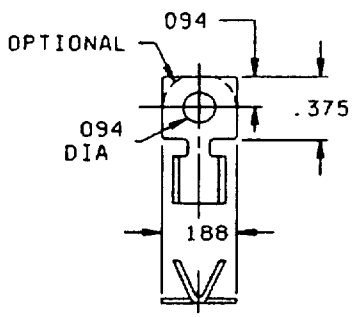
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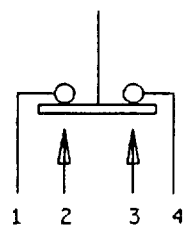
INTERNAL TOOTH LOCKWASHER



OPTIONAL MOUNTING NUT
SEE NOTE 7



SOLDER TERMINAL LUG



CIRCUIT SCHEMATIC

Inches	mm	Inches	mm	Inches	mm
001	0 03	125	3 18	390	9 91
.005	0 13	187	4 75	500	12 70
008	0.20	188	4.78	515	13.08
012	0 30	220	5 59	562	14 27
015	0 38	250	6 35	625	15 88
022	0 56	281	7 14	68	17 27
031	0 79	312	7 92	1.250	31 75
.094	2 39	.375	9 53		

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ± 015
4. Exact shape of switch optional provided dimensions specified are not exceeded
5. Panel seal gasket supplied with enclosure design 6 (splashproof) switches
6. Recommended mounting hole 0.515 +.008/-.001 diameter
7. Optional mounting nut is MS25082-B22 or -C22, as applicable

FIGURE 1. Configuration and dimensions - Continued.

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REQUIREMENTS:

Dimensions and configuration See figure 1

Enclosure design 1 (unsealed), 2 (dusttight), or 6 (splashproof) as defined by Part or Identifying Number (PIN)

Temperature characteristic

MS16712-1 and -2. Room temperature only

All other PIN's 1 (-55°C to +85°C)

Shock type H (high-impact) No opening or closing of contacts shall exceed 20 ms duration.

Sinusoidal vibration grade 2 (10 to 2,000 Hz)

Weight. 0 020 pound approximately

Solderability. Applicable to screw terminals and solder lugs.

Operating characteristics

Actuating force 3 5 pounds ±1 pound

Total travel

MS16712-1 and -2. .085 inch (2 16 mm) approximately.

All other PIN's. .085 ± 015 inch

Strength of actuating means

MS16712-1 and -2 25 pounds

All other PIN's 50 pounds

Terminal strength 5 pounds

Material

Case Anodized aluminum

MS16712-1 and MS16712-2 Clear finish

All other PIN's. Black finish

Mounting nut.

MS16712-1 and -2 Clear finish

All other PIN's. Black finish.

Internal tooth lockwasher. Phosphor bronze

Push button May be glass filled nylon

Screw terminals. May be silver plated brass

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Dielectric withstanding voltage·

MS16712-1 and -2. In qualification inspection table after electrical endurance, the dielectric withstanding voltage points of application between terminals is not applicable

Mechanical endurance

MS16712-1 and -2 50,000 cycles minimum

All other PIN's 200,000 cycles minimum

Electrical ratings

Endurance testing: See table I

Part or Identifying Number (PIN) See table II.

TABLE I Electrical ratings (endurance testing)

Load	Volts	Amperes	Endurance cycles
Resistive	115 V ac, 400 Hz 28 V dc	10 10	25,000 at sea level 25,000 at sea level
Lamp	115 V ac, 400 Hz 28 V dc	3 3	25,000 at sea level 25,000 at sea level
Inductive in accordance with MIL-I-81023	115 V ac, 400 Hz 28 V dc	5 5	25,000 at sea level 5,000 at 50,000 feet
5 power factor inductive	120 V ac, 60 Hz	7 5	40,000 at sea level
Inductive, 2 henrys	230 V dc	0 05	5,000 at sea level

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TABLE II. PIN's and characteristics

PIN	Enclosure design	Button color 1/	Case and mounting nut finish	Temperature characteristic	Mechanical endurance cycles
MS16712-1	Unsealed	Black	Clear	Room temperature	50,000
MS16712-2	Unsealed	Red	Clear	Room temperature	50,000
M8805/20-11	Unsealed	Red	Black	1(-55°C to +85°C)	200,000
M8805/20-12	Unsealed	Black	Black	1(-55°C to +85°C)	200,000
M8805/20-13	Unsealed	Orange	Black	1(-55°C to +85°C)	200,000
M8805/20-14	Unsealed	Yellow	Black	1(-55°C to +85°C)	200,000
M8805/20-15	Unsealed	Green	Black	1(-55°C to +85°C)	200,000
M8805/20-16	Unsealed	Blue	Black	1(-55°C to +85°C)	200,000
M8805/20-17	Unsealed	Violet	Black	1(-55°C to +85°C)	200,000
M8805/20-18	Unsealed	Gray	Black	1(-55°C to +85°C)	200,000
M8805/20-19	Unsealed	White	Black	1(-55°C to +85°C)	200,000
M8805/20-21	Dusttight	Red	Black	1(-55°C to +85°C)	200,000
M8805/20-22	Dusttight	Black	Black	1(-55°C to +85°C)	200,000
M8805/20-23	Dusttight	Orange	Black	1(-55°C to +85°C)	200,000
M8805/20-24	Dusttight	Yellow	Black	1(-55°C to +85°C)	200,000
M8805/20-25	Dusttight	Green	Black	1(-55°C to +85°C)	200,000
M8805/20-26	Dusttight	Blue	Black	1(-55°C to +85°C)	200,000
M8805/20-27	Dusttight	Violet	Black	1(-55°C to +85°C)	200,000
M8805/20-28	Dusttight	Gray	Black	1(-55°C to +85°C)	200,000
M8805/20-29	Dusttight	White	Black	1(-55°C to +85°C)	200,000
M8805/20-61	Splashproof	Red	Black	1(-55°C to +85°C)	200,000
M8805/20-62	Splashproof	Black	Black	1(-55°C to +85°C)	200,000
M8805/20-63	Splashproof	Orange	Black	1(-55°C to +85°C)	200,000
M8805/20-64	Splashproof	Yellow	Black	1(-55°C to +85°C)	200,000
M8805/20-65	Splashproof	Green	Black	1(-55°C to +85°C)	200,000
M8805/20-66	Splashproof	Blue	Black	1(-55°C to +85°C)	200,000
M8805/20-67	Splashproof	Violet	Black	1(-55°C to +85°C)	200,000
M8805/20-68	Splashproof	Gray	Black	1(-55°C to +85°C)	200,000
M8805/20-69	Splashproof	White	Black	1(-55°C to +85°C)	200,000

1/ Button colors shall be in accordance with FED-STD-595 as follows Red-11105, black-17038, orange-12510, yellow-13655, green-14090, blue-15102, violet-17142, gray-16493, white-17886

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Qualification inspection.

Group submission: See table III

TABLE III Group submission qualification inspection

Inspection	Sample	Extent of Approval
Group I	All sample units	All
Group II	Unsealed, temperature characteristic 1 2 samples Splashproof, temperature characteristic 1. 2 samples	
Group IV	Unsealed 2 samples	
Group V	Unsealed 2 samples	
Group VI	Dusttight, temperature characteristic 1 4 samples	
Group VII	Dusttight, temperature characteristic 1 10 samples	
Group VIII	Dusttight 1 sample Splashproof 1 sample	

Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AV
Navy - AS, OS
Air Force - 99
DLA - ES

User activities
Army - AR
Navy - MC
Air Force - 19

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5930-1499)

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